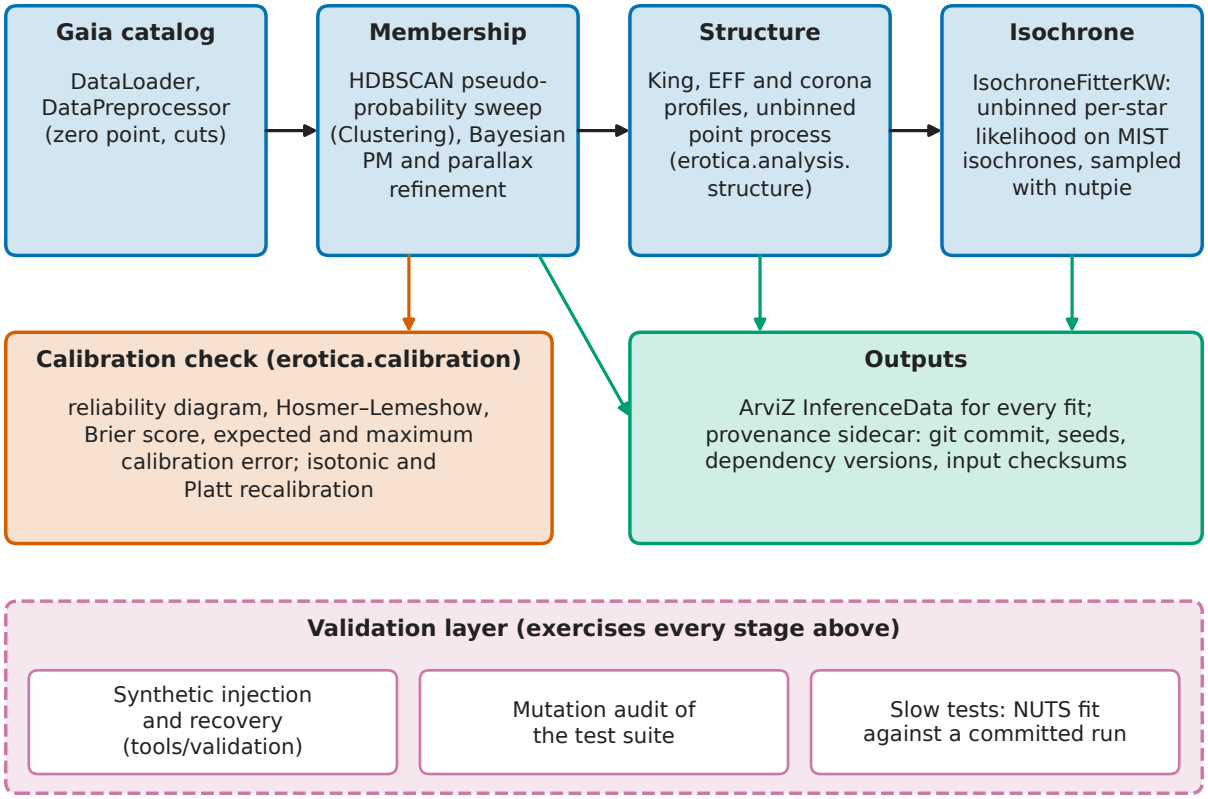
Gaia catalog
DataLoader, DataPreprocessor (zero point, cuts)
Membership
HDBSCAN pseudo-probability sweep (Clustering), Bayesian PM and parallax refinement
Structure
King, EFF and corona profiles, unbinned point process (erotica.analysis. structure)
Isochrone
IsochroneFitterKW: unbinned per-star likelihood on MIST isochrones, sampled with nutpie
Calibration check (erotica.calibration)
reliability diagram, Hosmer–Lemeshow, Brier score, expected and maximum calibration error; isotonic and Platt recalibration
Outputs
ArviZ InferenceData for every fit; provenance sidecar: git commit, seeds, dependency versions, input checksums
Validation layer (exercises every stage above)
Synthetic injection and recovery (tools/validation)
Mutation audit of the test suite
Slow tests: NUTS fit against a committed run